what is data science?

Data science ek technique ha jis ma hum data collect,analyze or understand karta ha ta ka data sa useful information or decision nekala ja saka ya data scienca kahlata ha.

- Environment science ka department hama batata ha ka ic din barish ho ge ya ic din badil ayya ga ya taz hava aya ge flood ka masla ho ga ya sab data science ma ata ha.
- Busness developement mesal ka tor par Amazon,Google in ko busnass egencies menage karti ha hum na koi cheas buy karni ha hum google par search karta ha to us ka adds facebook ,tiktok par ana shro ho gata ha ya sab data science model ha.

What is bioinformatics?

Bioinformatics sa murad ham computer or sofware ka use kar ka biology information (hum biology or computer science dono ko mila kar data)ko store ,analyze or understand karta ha.

- For example gab koi vacean bani gati ha to us ka bara ma bataya gata ha ka ic ka 90 percent side effect ha ya ic ma ya ya genes ka istamal kaya gaya.

> Python sab sa zayada use hona vali programming language ha.

Python ko bahat sa log use karta ha ya Programmer,Data analyst,Matamaticions,Acountent,Network engineers or every body can use python.

> Python is a open source programming language.

Puthon ke madad sa ham same task ko bar bar repeat kar sakta ha.

Lacture -2

> Ten important steps to become a data scientist.

1-Laran basic matamatics

2-Laran basic statistics

3-Learn a programming language (Python or R)

4-Data wrangling and data cleaning

5-Data visualization

6-Machine/ Deep learning Alogrithms

7-Linux,Coda ,Git or enviornmnt

8-Open source database

9-Domain sapecific,Data science or project.

10-Motivation and Good PC

Lacture -3 (Day 1a)

Jab ham data science par kam karta ha to python ka use hota ha ic leya ham data science sa pahla python ko explain kara ga.Phla lacture ma hum data science ka bara ham dic kas kar chokaa ha.

- Data science ek field ha gs ma hum data collect,analyze or understand karta ha ta ka data sa useful informatin banya ga sakaa or decision nekala ja saka.

Machine learning sa kaya murad ha?

Machine learning artficial intelligence ka ak component hota ha ic ma computer ma data or exemple deye gati haa or computer in example sa pattern sekta ha or fature ma decision lana ke koshsh karta ha.

Example

Youtube apke watch hitory or dosra signl ko analyze kar ka andaza laga sakta ha ka ap ko kon si video passand aa sakti haa

Lacture -4(Day 1)

Sab sa pahla ham na window ma python ko run kasa karna ha?

pahla hama python ko install karna para ga.phir ham apni window par command prompt ko click kara ga or write kara ga python --version or enter kara ga agar to python ka version show ho gya to python installed ha nahi to hama install kari para ge

Agar download nahi ha to ham google par search kara ga **python.org** or yaha sa window par ga kar ham download kara ga.

Ab hum dobara sa search kara ga **Python --version** to apna version show karvaa ge .or python sa bahir gana ka laya hum exit() write kara ga.



Ic ka alva hum search bar python search kara ga to install python ka terminal open ho gya ga or jo be command write kara ga vo python ma open ho ge

python ko us ke lenguage ka elava koi instruction de gya to vo usa perfom nahi karta

- Example

Agar hum python ko kahta ha ka **What is 2+3** to vo ic ka koi javab nahi da ge ic ka elava hum ic ko directly khta ha ka 2+3 to vo foran us ka javab da ge.

> Python ko use karna ka method: print ka p small or brakts ma vo word write karna go display karvana ho US ko colon ma likhna ha.

- Example

print("Hello World")

Hello Worid

Python case sencitive language ha agar print ka p capital lakha ga to ya execute nahi kara ga

Ic ka bad hum simply disktop par ak text file create karta ha ic ma be ham apni commands write kar sakta ha go python understand kar saka.

 alt text

Agar ham cammond prompt par write kara ga *cd Desktop* to desktop par chala gya ga phir yaha ham na go folder banaya ha *cd adil_javed* ka nam sa likha ga to vo folder open ho gya ga,agar open folder ka back gana ha to ham us ka aga cd double dot (cd ..) likha ga to vo folder vapis ho gya ga.

- Agar ham kese folder ka samana code single . (code .) write karta ha to to folder VS code ma open ho ga.
- Agar ham na koi folder banana ha to ham cd Desktop write kara ga next **mkdir** write kar ka phir go folder ko nam dana ha vo write kara ga (mkdir adil_javed) to folder desktop par ban gaya ga.
- Agar hm command prompt par *dir* write kara ga to vo desktop ka all folders ko show karva da ga

 alt text

Phir ham VS code ko instal karta ha or ic ma python ke extension instal karta ha ya be python ko chlana aa ka laya bahter cerdar ada karta ha VS code ma hum ak file banata ha us ko ham 01_first_program.py ka nam dataa ha jasa he ham .py write karta ha to file open ho gya ge.

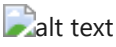
Ic python ke file ma ak program lakhaa ga go hama terminal ma show karvaa ga

 alt text

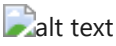
Lacture -5 (Day 2)

Ic ma ham operater ka use ka bara ma parra ga ko vo pythn ma kasa kam karta ha.

Ham VS code ma ak new file create kara ga jis ko ham **01_operater.py** ka nam da ga. phir simply ham 2+3,3-2,2*3,8/4 or 9/4 to vo in ka correct answer da ga. ham 9//4 write kara ga to error da ga


alt text

Python ma gab ham operater ko use karta haa to spaces ka koi issue nahi hota


alt text

Above picture ma ham na spaces be deye ha lakin answer corect aya ha.

Agar ham code ko comments ma change kar da ga to vo run / execute nahi ho ga yani code ka sat hash # laga da ga to vo code run nahi ho ga.agla code par chala gya ga

python PEMDAS ke sequance ma kam karta ha .

> PEMADS stand for Parenthesis Exponents Multiply Devide Addition Sebstraction.
>
> python ma code ke execution left to right ma hoti ha yani M D A S.